# 1 Linear Systems and Stability

Consider the continuous-time linear time-invariant system $\dot{x}(t) = Ax(t) + Bu(t)$, where $A \in \mathbb{R}^{n \times n}$. The system is asymptotically stable if and only if every eigenvalue $\lambda_i$ of $A$ satisfies $\mathrm{Re}(\lambda_i) < 0$.

$$\int_0^\infty e^{A^\top t} Q\, e^{At}\, dt = P, \qquad A^\top P + PA = -Q. \tag{1}$$

The transfer function is $G(s) = C(sI - A)^{-1}B + D$, with poles at the roots of $\det(sI - A) = 0$. This sentence is set in 50% gray, as used for captions and side notes in many textbooks. Hyperlink-blue text, a red warning, and green remark.

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u, \quad \sum_{k=1}^{N} \frac{1}{k^2} \approx \frac{\pi^2}{6}. \tag{2}$$

Small text, footnote size: the Lyapunov function $V(x) = x^\top P x$ is positive definite and decreases along trajectories.

Highlighted box with black text on a pale yellow fill, common in lecture slides.